Viability-Preserving Rollout Approximate Dynamic Programming for Required-Target Orbital Inspection

Rugang Tang
Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong SAR

Zheng Tan
Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong SAR

Wenyu Yang
Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong SAR

Xin Ning
School of Astronautics, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, China

Chih-Yung Wen
Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong SAR

***Abstract*— Autonomous orbital inspection requires maneuver-efficient completion of prescribed target observations under motion and safety constraints. This paper proposes viability-preserving rollout approximate dynamic programming (ADP) on a directed graph of audited orbital arcs. Finite-depth Bellman lookahead evaluates admissible sequences using complete task-aware base-policy continuations. Actions lacking a certified task-completing continuation receive infinite value, while admissible connections with no immediate inspection gain remain available. The algorithm selects the minimum-value first action and replans after each observation. Under deterministic finite-graph assumptions, exhaustive rollout preserves base-policy task completion without increasing graph cost. A separate depth diagnostic supports retaining depth three to balance cost and computation. The method completes every mission in an independently certified all-solvable development benchmark, with shorter median runtimes than an independent mixed-integer linear program. In a separate fixed-library confirmation, mean maneuver velocity increment decreases by 11.05% relative to one-step-ADP-seeded local search on jointly completed test missions. A Robot Operating System 2 (ROS 2) simulation verifies fixed-route tracking and required-target completion under the enabled safety checks. These results support completion-certified inspection sequencing within the audited finite-graph model.**

This work was supported in part by the Research Centre for Unmanned Autonomous Systems of The Hong Kong Polytechnic University under Grant P0046487. *(Corresponding author: Zheng Tan)* (e-mail: zheng-uav.tan@connect.polyu.hk).

0018-9251 © IEEE

***Index Terms*— spacecraft inspection, approximate dynamic programming, safe observable orbital arc, relative orbital dynamics, inspection requirements**

I. Introduction

Orbital inspection provides visual evidence for spacecraft assessment and on-orbit servicing [1]. Robot-aided inspection was demonstrated during STS-85 [2], and later planning studies address large structures with one or several free-flying inspectors [3]–[5]. An autonomous inspector must select observation locations and their order because relative dynamics couple transfer feasibility and maneuver effort.

Next-best-view methods model observation quality [6], kinodynamic planners combine coverage and differential constraints [3], [4], and relative-orbit planners exploit natural motion [7], [8]. Prescribed missions require specific targets, which an aggregate coverage threshold can miss. We therefore select candidate viewpoints and optimize their sequence to observe every required item.

The planning challenge is to reduce cumulative maneuver effort while preserving a feasible continuation to all required observations. A low-cost immediate transfer can leave costly remaining observations or exhaust the budget before completing the task. Each safe observable orbital arc (SOOA) combines a source-dependent Hill–Clohessy–Wiltshire (HCW) transfer, a stabilized camera observation, and feasibility audits. Safe connecting arcs remain useful even when they reveal no new target immediately. Fig. 1 presents the resulting workflow, from mission inputs and arc construction to rollout ADP and inspection records.

The main contribution is a viability-preserving rollout approximate dynamic programming (ADP) formulation for maneuver-efficient required-target inspection. It specializes established rollout policy improvement [9], [10] using complete task-aware base-policy evaluations and finite-depth Bellman expansion. Actions are evaluated by approximate total graph cost; those lacking a certified task-completing continuation receive infinite value. The analysis establishes completion preservation and base-policy no-degradation for SOOA sequencing, with exact dynamic programming providing a reference on tractable graphs.

The contributions are:

1) a viability-preserving rollout ADP over directed SOOA graphs for maneuver-efficient required-target inspection, combining required-target terminal predicates, finite-depth Bellman improvement, complete task-aware base-policy evaluation, audited visibility and motion transitions, and safe zero-gain connectors;
2) a finite-graph analysis proving completion preservation and base-policy cost non-degradation for repeated SOOA rollout, with exact dynamic programming providing a reference on exhaustible reduced graphs; and

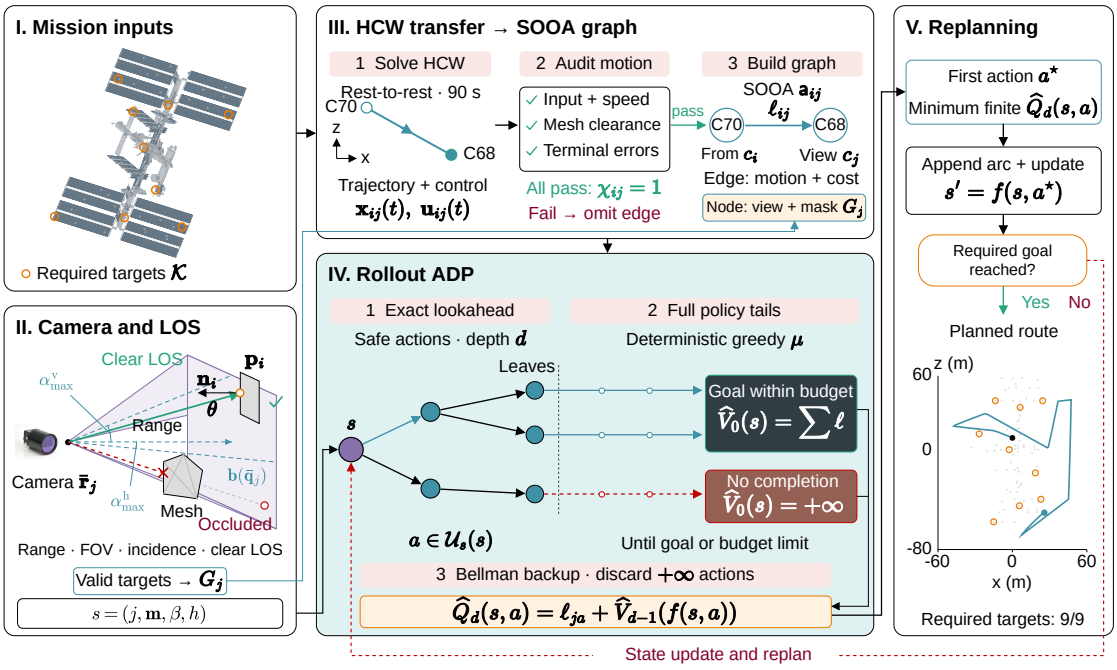


Fig. 1. Viability-preserving rollout ADP for orbital inspection. I. International Space Station (ISS) mesh and required targets $\mathcal{K}$. II. Camera-valid targets G_j from range, field of view (FOV), incidence $\theta \le \theta_{\max}$, and line of sight (LOS) checks. III. HCW transfer, motion audits, and directed SOOA construction; C70 and C68 are archived candidate IDs. IV. Depth-d Bellman lookahead with complete greedy base-policy tails. At each leaf, $\sum \ell$ denotes the full tail cost; $+\infty$ denotes no certified completion within the remaining budget. V. First-action selection and planning-state updates toward completion of all required targets. The tree is schematic; the 9/9 route is an archived offline plan. ISS geometry: National Aeronautics and Space Administration (NASA), NASA.

3) a reproducible evaluation of task completion, maneuver efficiency, and execution feasibility. The independently certified all-solvable benchmark separates planner performance from viewpoint availability, while visibility-infeasible cases form a separate stress test. Fixed-library comparisons quantify maneuver savings, and Robot Operating System 2 (ROS 2) execution checks verify reference tracking and required-target completion.

Notation: $\mathbb{R}^n$ denotes the space of real n-vectors, bold lowercase symbols denote vectors, and $(\cdot)^\mathsf{T}$ and $\|\cdot\|$ denote transpose and Euclidean norm. For a set $\mathcal{S}$, $|\mathcal{S}|$ is its cardinality; $\mathrm{supp}(\mathbf{m}) = \{i : m_i = 1\}$ is the support of a binary vector. The scalar indicator $\mathbf{1}\{\cdot\}$ equals one when its argument is true and zero otherwise, $\mathbf{1}_G$ is the indicator vector of set G, and $\vee$ denotes elementwise logical OR. Finally, $\mathbf{0}$ and I_n denote a zero vector and the $n \times n$ identity matrix, respectively.

II. Related Work

Orbital inspection and sensing: Inspection connects spacecraft observation to servicing decisions [1], [2]. Vision-based navigation has also been tested aboard the International Space Station (ISS) with the VERTIGO/SPHERES facility [11]. That intravehicular microgravity testbed provides experimental sensing evidence, distinct from external structure inspection. For external missions, camera visibility and collision avoidance must be considered together: Wang et al. combine position and attitude constraints in barrier-function-based inspection trajectory generation and tracking [12]. These requirements motivate the separation of observation acceptance, transfer admissibility, and task completion in OrbInspect.

Relative motion and inspection planning: The Clohessy–Wiltshire equations describe relative motion near a circular chief orbit [13], [14]. Williams derives inspection transfers using line-of-sight switching maneuvers [15]; Hou et al. design relative orbits for visual inspection of the China Space Station [7]; and Zhou et al. combine near-natural relative ellipses with formation control around a rotating station [16]. These approaches exploit orbital geometry and natural motion, which also determine the source-dependent costs and admissibility of the SOOA edges.

Robotic view planning couples sensor range, incidence, occlusion, and reconstruction quality to motion cost [6]. Rapidly exploring random trees and asymptotically optimal sampling methods provide motion-search foundations [17], [18]. Spacecraft inspection planners extend these ideas to kinodynamic coverage [3], [4], receding-horizon inspection of unknown noncooperative objects [19], and astrodynamics-informed transfer sampling [8]. Recent methods further address coverage–energy trade-offs [5] and surface-aware tomographic paths [20]. The present decision problem instead fixes a candidate library and prescribed target identities, then optimizes their completion sequence over audited transfers.

ADP and learning-based spacecraft control: ADP approximates expensive exact cost-to-go calculations [10], [21], while combinatorial rollout improves sequential decisions through heuristic completions [9]. Its policy-iteration interpretation connects rollout with model predictive control and terminal-set optimization [22], [23]. Learning-based spacecraft applications include proximity control [24], [25], online attitude stabilization [26], reach-avoid games [27], and run-time-assured inspection [28]. OrbInspect instead uses complete model-based base-policy simulations to certify finite observation sequences.

Safety and completion guarantees: Constrained policy optimization regulates expected constraint costs [29], whereas shielding restricts action selection using a specified safety model [30]. Control barrier functions impose conditions for forward invariance [31], and predictive barrier functions provide a recovery mechanism toward the feasible set of a predictive safety filter [32]. OrbInspect separates these control-level concerns from finite-task viability: motion audits admit SOOA edges, while finite rollout values certify a continuation to every required target. The analysis specializes established rollout principles to establish finite-task completion and graph-cost bounds for sequencing audited inspection arcs.

III. System and Observation Model

This section defines the deterministic system, observation, and feasibility model from which the candidate graph is constructed. The inspected structure is represented by a known station mesh and surface samples; the chaser state, camera parameters, nonempty required-target set, keep-out model, and actuation limits define each scenario. These ingredients determine visibility masks, admissible transfers, and SOOA edge costs.

Table I summarizes the modeling assumptions and their implications.

A. Relative Dynamics

The inspected structure is fixed in a local-vertical/local-horizontal (LVLH) frame attached to a nominal circular chief orbit. Its axes are radial, along-track, and cross-track, respectively. The chaser state is $\mathbf{x} = [\mathbf{r}^\mathsf{T}, \mathbf{v}^\mathsf{T}]^\mathsf{T} \in \mathbb{R}^6$, where $\mathbf{r}, \mathbf{v} \in \mathbb{R}^3$ are the relative position and velocity. With mean motion n and commanded acceleration $\mathbf{u} \in \mathbb{R}^3$, the HCW model is

$$\begin{aligned} \dot{\mathbf{x}} &= A_{\mathrm{HCW}}\mathbf{x} + B\mathbf{u}, \\ A_{\mathrm{HCW}} &= \begin{bmatrix} 0 & I_3 \\ K_n & C_n \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ I_3 \end{bmatrix}, \\ K_n &= \mathrm{diag}(3n^2, 0, -n^2), \quad C_n = \begin{bmatrix} 0 & 2n & 0 \\ -2n & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned} \tag{1}$$

With piecewise-constant acceleration over Δt, fixed-step fourth-order Runge–Kutta (RK4) integration induces the linear discrete map

$$\mathbf{x}_{k+1} = \widetilde{A}_d\mathbf{x}_k + \widetilde{B}_d\mathbf{u}_k, \tag{2}$$

where $\widetilde{A}_d \in \mathbb{R}^{6\times6}$ and $\widetilde{B}_d \in \mathbb{R}^{6\times3}$ are the RK4 state and input matrices associated with (1). The implementation evaluates the RK4 stages directly.

B. Visibility, Prescribed Observation, and Coverage

Let $\mathcal{T} = \{1, \dots, N\}$ denote the sample-index set. The structure is represented by indexed surface samples $\{(\mathbf{p}_i, \mathbf{n}_i, \Delta A_i)\}_{i=1}^N$, where $\mathbf{p}_i \in \mathbb{R}^3$ is the sample position in the LVLH frame, $\mathbf{n}_i \in \mathbb{R}^3$ is its outward unit normal, and $\Delta A_i > 0$ is its represented surface area. Mission priority weights are separately defined as $w_i = \Delta A_i \eta_i$, where $\eta_i > 0$ is a priority multiplier. A fixed nonempty set $\mathcal{K} \subseteq \mathcal{T}$ specifies the required inspection items before scenario evaluation.

The camera attitude quaternion $\mathbf{q}_k$ specifies unit boresight $\mathbf{b}(\mathbf{q}_k)$ and the horizontal and vertical camera axes. Let $d^{\mathrm{f}}_{i,k}$, $d^{\mathrm{h}}_{i,k}$, and $d^{\mathrm{v}}_{i,k}$ denote the forward, horizontal, and vertical components of $\mathbf{p}_i - \mathbf{r}_k$ in this camera frame. With $\mathcal{O}$ denoting the occluding mesh, target i is visible when $d^{\mathrm{f}}_{i,k} > 0$ and

$$\begin{aligned} r^{\mathrm{cam}}_{\min} \le \|\mathbf{p}_i - \mathbf{r}_k\| &\le r^{\mathrm{cam}}_{\max}, \\ |d^{\mathrm{h}}_{i,k}| &\le d^{\mathrm{f}}_{i,k} \tan\alpha^{\mathrm{h}}_{\max}, \\ |d^{\mathrm{v}}_{i,k}| &\le d^{\mathrm{f}}_{i,k} \tan\alpha^{\mathrm{v}}_{\max}, \\ \arccos\frac{(\mathbf{r}_k - \mathbf{p}_i)^\mathsf{T}\mathbf{n}_i}{\|\mathbf{r}_k - \mathbf{p}_i\|\,\|\mathbf{n}_i\|} &\le \theta_{\max}, \\ \mathrm{LOS}(\mathbf{r}_k, \mathbf{p}_i, \mathcal{O}) &= 1. \end{aligned} \tag{3}$$

Here $r^{\mathrm{cam}}_{\min}$ and $r^{\mathrm{cam}}_{\max}$ are the camera range limits, and $\alpha^{\mathrm{h}}_{\max}, \alpha^{\mathrm{v}}_{\max} \in (0, \pi/2)$ are the horizontal and vertical FOV half-angles. The incidence-angle limit is $\theta_{\max}$, and $\mathrm{LOS}(\mathbf{r}_k, \mathbf{p}_i, \mathcal{O}) = 1$ denotes an unobstructed LOS. Let $\mathbf{m}_k \in \{0,1\}^N$ denote the cumulative covered-target mask after the first k selected observations, initialized at $\mathbf{m}_0 = \mathbf{0}$. Let M denote the number of candidate observation nodes, indexed by $j = 1, \dots, M$. For a target-index set G_j, let $\mathbf{1}_{G_j} \in \{0,1\}^N$ denote its indicator vector; let $\mathbf{e}_a \in \{0,1\}^M$ denote the standard basis vector that marks selected destination a. Fig. 1, panel II, illustrates the rectangular FOV and the range, incidence, and LOS tests. LOS is queried against a deterministic bounding-volume hierarchy containing every ISS mesh triangle; intersections at the target endpoint are excluded, while interior segment intersections block the view. Applying (3) at the candidate pose $(\mathbf{r}_k, \mathbf{q}_k) = (\bar{\mathbf{r}}_j, \bar{\mathbf{q}}_j)$ yields its fixed visibility mask G_j. Selecting node j credits this mask, so $\mathbf{m}_{k+1} = \mathbf{m}_k \vee \mathbf{1}_{G_j}$. Here $\mathbf{m}_k = [m_{1,k}, \dots, m_{N,k}]^\mathsf{T}$, where $m_{i,k} = 1$ exactly when target/sample i has been credited after the first k observations. Candidate generation can make a subset of the sampled surface invisible from every candidate observation. For the nonempty set $\mathcal{T}_{\mathrm{insp}} = \cup_{j=1}^M G_j$, the priority-weighted inspectable coverage is

$$C^{\mathrm{insp}}(\mathbf{m}_k) = \frac{\sum_{i\in\mathcal{T}_{\mathrm{insp}}} w_i m_{i,k}}{\sum_{i\in\mathcal{T}_{\mathrm{insp}}} w_i}. \tag{4}$$

TABLE I
Modeling assumptions and their implications

Assumption	Role in the baseline	Consequence for interpretation
HCW and fixed local-vertical/local-horizontal (LVLH) geometry	A circular chief orbit supports linear HCW relative translation, while a station fixed in LVLH keeps the mesh and safety geometry stationary during one planning run.	Quantitative results use a local linear, stationary-mesh model; perturbed ephemerides or station attitude motion require an extension.
Known mesh and state	Makes visibility, occlusion, and transfer evaluation deterministic.	Navigation, map, and shape uncertainty are not represented by the present safety margin.
Prescribed camera attitude	Points the boresight toward each candidate observation target and exports the corresponding quaternion stream.	Camera slew time, reaction-wheel limits, and closed-loop attitude error are outside the optimization.
Mesh-based sampled safety	Computes distance to the full International Space Station (ISS) triangle mesh at every rollout sample and rejects sampled motion segments that intersect the mesh.	Positive sampled clearance and surface nonintersection are planning audits rather than a robust continuous-time collision certificate.

The required-target coverage fraction and the binary required-target completion indicator are defined as

$$C^{\mathcal{K}}(\mathbf{m}) = \frac{|\mathcal{K} \cap \mathrm{supp}(\mathbf{m})|}{|\mathcal{K}|}, \qquad \Gamma_{\mathcal{K}}(\mathbf{m}) = \mathbf{1}\{\mathcal{K} \subseteq \mathrm{supp}(\mathbf{m})\}. \tag{5}$$

Completion denotes acceptance of the prescribed samples under (3). Graph-planning durations count transfers only, with exposure assumed complete at each stabilized candidate. The ROS execution durations additionally include terminal settling.

C. Dynamic Feasibility and SOOAs

Let r_{veh} denote the finite-body radius used to bound the chaser, and let d_{safe} denote the prescribed mesh-clearance margin. For an edge rollout with K post-step samples, every sampled state must satisfy

$$\|\mathbf{u}_k\|_2 \le u_{\max}, \qquad \|\mathbf{v}_k\|_2 \le v_{\max}, \qquad d_{\mathcal{S}}(\mathbf{r}_k) \ge 0, \tag{6}$$

where $d_{\mathcal{S}}(\mathbf{r}) = d_{\mathrm{mesh}}(\mathbf{r}) - r_{\mathrm{veh}} - d_{\mathrm{safe}}$ is finite-body clearance above the required margin, d_{mesh} is unsigned center distance to the closest triangle in the complete ISS mesh, and $d_{\mathcal{S}} = 0$ is the sampled admissibility boundary. The mesh coordinates are obtained from the complete GL Transmission Format (glTF) scene hierarchy, followed by the Simulation Description Format (SDF) pose and scale. In addition, each consecutive pair of rollout samples is tested for a triangle-surface intersection, and its continuous center-distance lower bound subtracts half the segment length from the smaller endpoint distance by the 1-Lipschitz property of distance to a closed set. This swept test catches a direct mesh crossing even if both endpoints have positive unsigned distance.

For a directed rollout from source node i to destination node j, the desired terminal position is $\bar{\mathbf{r}}_j$ and velocity is zero. Define the root-mean-square (RMS) distance to this fixed destination and the terminal position and velocity errors, respectively, as

$$e_{ij}^{\mathrm{rms}} = \left(\frac{1}{K} \sum_{k=1}^{K} \|\mathbf{r}_{ij,k} - \bar{\mathbf{r}}_j\|_2^2 \right)^{1/2}, \qquad e_{ij}^{r} = \|\mathbf{r}_{ij,K} - \bar{\mathbf{r}}_j\|_2, \qquad e_{ij}^{v} = \|\mathbf{v}_{ij,K}\|_2. \tag{7}$$

The RMS distance summarizes all K post-step samples and contributes to the graph cost. Terminal errors use only the final sample and enter the admissibility checks. Also define $\Delta v_{ij} = \sum_k \|\mathbf{u}_{ij,k}\|_2 \Delta t$ and $d_{ij}^{\min} = \min_k d_{\mathcal{S}}(\mathbf{r}_{ij,k})$. The fixed nonnegative weights λ_v, λ_e, λ_s, and λ_a correspond to maneuver increment, RMS distance to the destination, clearance violation, and per-edge action cost, respectively. The graph stage cost is

$$\ell_{ij} = \lambda_v \Delta v_{ij} + \lambda_e e_{ij}^{\mathrm{rms}} + \lambda_s \max(0, -d_{ij}^{\min}) + \lambda_a. \tag{8}$$

For any route, J denotes the sum of its edge stage costs; J_{rollout} in the illustrative figure is the corresponding realized rollout cost. For every shield-admissible edge, the clearance-violation term is zero. It remains in the shared cost implementation so deliberately unshielded comparison methods can be evaluated with the same expression.

A candidate observation node and a directed SOOA edge are, respectively,

$$c_j = (\bar{\mathbf{r}}_j, \bar{\mathbf{q}}_j, G_j), \qquad \mathsf{a}_{ij} = \big(\mathbf{x}_{ij}[1{:}K], \mathbf{u}_{ij}[0{:}K-1], \mathbf{b}_{ij}[1{:}K], \Delta v_{ij}, d_{ij}^{\min}, u_{ij}^{\max}, v_{ij}^{\max}, e_{ij}^{r}, e_{ij}^{v}, \rho_{ij}\big). \tag{9}$$

Here $\bar{\mathbf{q}}_j$ is the prescribed attitude quaternion, and $G_j \subseteq \mathcal{T}$ is the camera-valid target set. The terminal errors e_{ij}^{r} and e_{ij}^{v} are defined in (7). The stored state and boresight arrays contain the K post-step samples, while $u_{ij}^{\max}$ and $v_{ij}^{\max}$ record the maximum sampled input and speed norms; the source state is held by candidate node c_i. In the edge notation below, i and j index candidate nodes and are distinct from the surface-sample index used above. The initial condition is a distinguished source node c_0 with no visibility reward. For an optional passive audit over L_p steps, zero-input HCW drift can be propagated from every stored edge state and recorded as ρ_{ij} =

$\min_{1\le k\le K,\,0\le l\le L_p} d_{\mathcal{S}}(\mathbf{r}^{\mathrm{drift}}_{ij,k}(l))$. The reported experiments enable input, speed, full-mesh sampled-clearance, swept-intersection, and terminal checks; passive-drift auditing is disabled. Throughout this paper, "safe" means admissible under the audits explicitly enabled for the reported experiment.

The preceding model induces the required-target sequencing problem.

PROBLEM 1 (Required-target SOOA planning) *Given the candidate nodes and stored directed SOOA records above, an initial action budget H, and the required set $\mathcal{K}$, find a finite sequence $\pi = (a_1, \dots, a_L)$, with $0 \le L \le H$, that minimizes*

$$J(\pi) = \sum_{k=1}^{L} \ell_{j_{k-1}a_k}, \qquad j_0 = 0, \quad j_k = a_k, \tag{10}$$

subject to the following conditions: each traversed record $\mathsf{a}_{j_{k-1}a_k}$ passes all enabled motion, terminal, clearance, and passive audits; no destination is selected more than once; and the masks updated by $\mathbf{m}_k = \mathbf{m}_{k-1} \vee \mathbf{1}_{G_{a_k}}$, $\mathbf{m}_0 = \mathbf{0}$, satisfy $\Gamma_{\mathcal{K}}(\mathbf{m}_L) = 1$. A zero-gain action is allowed when it is needed to reach a later required observation. If no such sequence exists within H, the instance is declared infeasible in the available candidate graph.

IV. Viability-Preserving Rollout ADP

A. Candidate Graph and Markov State

The candidate graph contains the initial node c_0 and the M candidate observation nodes $c_1, \dots, c_M$ in (9). A directed edge from c_i to c_j is the source-dependent SOOA a_{ij}, and it is absent when an enabled audit fails. Thus visibility G_j is node dependent, whereas trajectory, cost, clearance, input, terminal error, and passive margin are edge dependent. This separation is necessary because reaching the same observation from two different source nodes generally produces different HCW motion and audit values.

A node identifier is not a sufficient decision state because the value of a view changes after its targets have been covered. The finite-horizon Markov state is

$$s_k = (j_k, \mathbf{m}_k, \boldsymbol{\beta}_k, h_k), \tag{11}$$

where $j_k \in \{0, \dots, M\}$ is the current node, $\mathbf{m}_k \in \{0,1\}^N$ is the covered-target mask, $\boldsymbol{\beta}_k \in \{0,1\}^M$ is the selected-observation mask, and h_k is the remaining action budget. Selecting action $a \in \{1, \dots, M\}$ chooses destination node c_a, traverses edge $\mathsf{a}_{j_k a}$, and gives the deterministic transition

$$f(s_k, a) = (a,\ \mathbf{m}_k \vee \mathbf{1}_{G_a},\ \boldsymbol{\beta}_k \vee \mathbf{e}_a,\ h_k - 1)\,. \tag{12}$$

The required set $\mathcal{K}$ is fixed problem data. The shield-admissible action set $\mathcal{U}_s(s)$ is defined later. The constrained graph problem minimizes total stage cost subject to (5), with zero terminal cost for a completed task and value $+\infty$ when no admissible completion exists within the remaining budget. Its Bellman recursion is

$$\begin{aligned} V^\star(s) &= \begin{cases} 0, & \Gamma_{\mathcal{K}}(\mathbf{m}) = 1, \\ +\infty, & h = 0 \text{ or } \mathcal{U}_s(s) = \varnothing, \\ \min\limits_{a \in \mathcal{U}_s(s)} Q^\star(s,a), & \text{otherwise}, \end{cases} \\ Q^\star(s,a) &= \ell_{ja} + V^\star(f(s,a)). \end{aligned} \tag{13}$$

Here j and h are the current-node and remaining-budget components of s, and f is the deterministic state-transition map. Exact evaluation is exponential in the coverage and selected-node masks. The proposed method approximates this recursion with a short exact prefix and a computationally inexpensive, goal-aware terminal policy.

B. SOOA Construction and Safety Shield

Area-weighted targets are sampled from the mesh, and camera seeds are generated on offset shells around their outward normals. Each fixed visibility mask G_j is computed once from the terminal observation geometry. For a transition to $\bar{\mathbf{x}}_j = [\bar{\mathbf{r}}_j^{\mathsf{T}}, \mathbf{0}^{\mathsf{T}}]^{\mathsf{T}}$, the planner solves the minimum-energy piecewise-constant HCW boundary-value problem under the RK4 map in (2):

$$\begin{aligned} \min_{\mathbf{u}_0, \dots, \mathbf{u}_{K-1}} \quad & \sum_{k=0}^{K-1} \|\mathbf{u}_k\|_2^2 \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = \widetilde{A}_d \mathbf{x}_k + \widetilde{B}_d \mathbf{u}_k, \\ & \mathbf{x}_K = \bar{\mathbf{x}}_j. \end{aligned} \tag{14}$$

If the terminal influence matrix $\mathcal{B}_K = [\widetilde{A}_d^{K-1}\widetilde{B}_d \ \cdots \ \widetilde{B}_d] \in \mathbb{R}^{6\times 3K}$ has full row rank, stack the commands as $\mathbf{U}_{ij} = [\mathbf{u}_{ij,0}^{\mathsf{T}}, \dots, \mathbf{u}_{ij,K-1}^{\mathsf{T}}]^{\mathsf{T}}$. The minimum-norm command from source state $\mathbf{x}_i^{\mathrm{src}}$ is then $\mathbf{U}_{ij}^\star = \mathcal{B}_K^{\mathsf{T}}(\mathcal{B}_K\mathcal{B}_K^{\mathsf{T}})^{-1}(\bar{\mathbf{x}}_j - \widetilde{A}_d^K \mathbf{x}_i^{\mathrm{src}})$. The implementation computes this expression through the 6×6 Gram system and then propagates the requested input without clipping. Let ε_r and ε_v denote the terminal position and velocity tolerances, respectively. Then $\chi_{ij} = 1$ exactly when every stored edge sample satisfies (6), the enabled swept-segment audit reports no triangle intersection, $e_{ij}^r \le \varepsilon_r$, $e_{ij}^v \le \varepsilon_v$, and $\rho_{ij} \ge 0$ when the passive audit is enabled. The shield admits

$$\mathcal{U}_s(s) = \{a \in \{1, \dots, M\} : h > 0,\ \beta_a = 0,\ \chi_{ja} = 1\}, \tag{15}$$

where j is the current-node component of s. Safe unvisited nodes remain admissible when they provide no immediate inspection gain, because a connecting action may be needed to reach a required target on the directed graph. The finite budget and no-revisit mask bound such connections. The shield makes an inadmissible edge unavailable to incumbent construction, local search, and exact recursion [30]. Shield admissibility is evaluated under the deterministic model using the enabled sampled-state, swept-intersection, and terminal checks.

An edge is audit-admissible when $\chi_{ij} = 1$; an action is shield-admissible when it belongs to $\mathcal{U}_s(s)$. A scenario

is feasible for the candidate graph when at least one shield-admissible sequence reaches $\Gamma_{\mathcal{K}}(\mathbf{m}) = 1$ within the available budget, and is infeasible when no such sequence exists.

C. Adaptive Base Policy and Rollout Value

Let $g_{\mathcal{K}}(s,a) = |\mathcal{K} \cap (G_a \setminus \mathrm{supp}(\mathbf{m}))|$ count newly accepted required targets, and let $\epsilon > 0$ prevent division by a near-zero stage cost. For the required-target task, the deterministic adaptive base policy is

$$\mu(s) \in \arg\max_{a \in \mathcal{U}_s(s)} \left(\frac{g_{\mathcal{K}}(s,a)}{\max\{\epsilon, \ell_{ja}\}}, g_{\mathcal{K}}(s,a), -\ell_{ja} \right), \quad (16)$$

where the tuple is compared lexicographically, followed by deterministic node-identifier tie breaking. The policy recomputes gain after every accepted observation; zero-gain ties favor lower stage cost. Starting from state s, its completion value is

$$\widehat{V}_0(s) = \begin{cases} \sum_{k=0}^{L-1} \ell_{j_k \mu(s_k)}, & \mu \text{ reaches} \\ & \Gamma_{\mathcal{K}}(\mathbf{m}) = 1 \text{ within } h, \\ +\infty, & \text{otherwise.} \end{cases} \quad (17)$$

where L is the number of actions taken by the base policy before completion or budget exhaustion. Thus $\widehat{V}_0$ approximates the cost-to-go through a complete, shielded base-policy simulation.

For rollout depth $d \geq 1$, define the truncated Bellman value

$$\widehat{V}_d(s) = \begin{cases} 0, & \Gamma_{\mathcal{K}}(\mathbf{m}) = 1, \\ +\infty, & h = 0, \\ \min\limits_{a \in \mathcal{U}_s(s)} \widehat{Q}_d(s,a), & \text{otherwise,} \end{cases} \quad (18)$$

where $\widehat{Q}_d(s,a) = \ell_{ja} + \widehat{V}_{d-1}(f(s,a))$. The minimum over an empty action set is $+\infty$. The policy selects a finite-valued minimizing action, applies (12), and rebuilds the depth-d tree at the successor state. State–depth memoization removes duplicate subproblems within each decision. The guarantees below require exhaustive prefix evaluation and the same deterministic Markov base policy at every leaf. Alg. 1 summarizes this finite-value action selection and repeated replanning procedure. Fig. 2 gives a worked decision on an illustrative graph, separating motion admissibility from completion certification and showing the continuation retained after replanning.

D. Rollout Guarantees and Exact Oracle

For fixed masks and nonnegative weights, $F(S) = \sum_{i=1}^{N} w_i \mathbf{1}\{i \in \cup_{j \in S} G_j\}$ is normalized, monotone, and submodular [33]. Unit weights on $\mathcal{K}$ give the same diminishing-returns property for required-target gain. This coverage fact neither justifies removing zero-gain connectors nor supplies a greedy approximation ratio, because SOOA cost and next-action admissibility depend on the current node.

Under the fixed deterministic graph assumptions, the state in (11) is Markov: j_k identifies the source-dependent

Algorithm 1 Viability-preserving rollout ADP

Require: Candidate graph, shield $\mathcal{U}_s$, required set $\mathcal{K}$, budget H, depth d

$s \leftarrow (0, \mathbf{0}, \mathbf{0}, H)$; $\pi \leftarrow ()$ ▷ π stores the selected action sequence
while $\Gamma_{\mathcal{K}}(\mathbf{m}) = 0$ and $h > 0$ **do**
  Enumerate every $a \in \mathcal{U}_s(s)$
  for each admissible a **do**
    $\widehat{Q}_d(s,a) \leftarrow \ell_{ja} + \widehat{V}_{d-1}(f(s,a))$
  end for
  Discard every action with $\widehat{Q}_d(s,a) = +\infty$
  if no finite action remains **then**
    return no certified completion
  end if
  $a^\star \leftarrow \arg\min_a \widehat{Q}_d(s,a)$ with deterministic ties
  Append $a^\star$ to π
  $s \leftarrow f(s,a^\star)$
end while
return π if $\Gamma_{\mathcal{K}}(\mathbf{m}) = 1$; otherwise return no certified completion

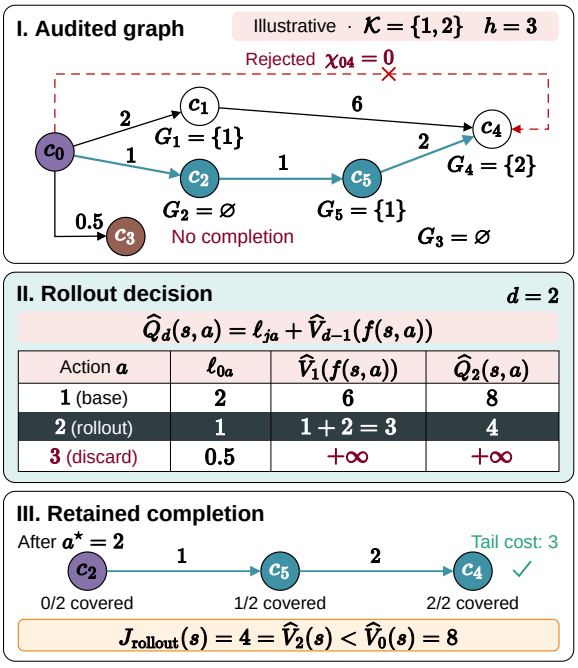


Fig. 2. Illustrative completion-certified rollout decision. I. The dashed edge is rejected by the shield. II. At $d = 2$, rollout selects the zero-gain connector c_2, with graph cost 4 rather than the base-policy cost 8; action 3 has no certified completion. III. Replanning at c_2 retains the cost-3 continuation. Values are illustrative; the experiments use $d = 3$.

edges, $\mathbf{m}_k$ determines unmet requirements, $\boldsymbol{\beta}_k$ enforces no revisits, and h_k gives the remaining budget. Partial dwell, independent look directions, or time-varying illumination would require additional state. If idealized rest-to-rest endpoints match exactly, concatenating edges with $\chi_{ij} = 1$ also preserves their recorded input, speed, sampled-clearance, swept-intersection, terminal, and enabled passive-drift results.

THEOREM 1 (Completion and base-policy bound) *Consider the deterministic finite graph, exhaustive depth-d prefix evaluation, and the same deterministic Markov base policy at every leaf. For fixed $d \geq 1$: (i) if $\widehat{V}_d(s) < +\infty$, repeated replanning in Algorithm 1 reaches the goal through shield-admissible actions within the remaining budget; and (ii) if $\widehat{V}_0(s) < +\infty$, then the realized graph cost satisfies $J_{\mathrm{rollout}} \leq \widehat{V}_d(s) \leq \widehat{V}_0(s)$.*

Proof:
Finiteness of $\widehat{V}_d(s)$ provides a shield-admissible depth-d prefix followed by a base-policy completion. After the first action, the retained tail, extended when necessary by the next base-policy action, remains a candidate in the rebuilt depth-d tree. The rebuilt value at the successor is therefore finite, and induction over the decreasing budget proves completion. If $\widehat{V}_0(s)$ is finite, the first d base-policy actions followed by the remaining base completion are feasible in (18), so $\widehat{V}_d(s) \leq \widehat{V}_0(s)$. More generally, whenever $\widehat{V}_{d-1}(s') < +\infty$, the same certificate can be extended by the next base-policy action when needed (or terminates earlier at the goal), giving $\widehat{V}_d(s') \leq \widehat{V}_{d-1}(s')$. Therefore, for a minimizing first action a and successor s', $\ell_{ja} + \widehat{V}_d(s') \leq \ell_{ja} + \widehat{V}_{d-1}(s') = \widehat{V}_d(s)$. Summing over replans yields $J_{\mathrm{rollout}} \leq \widehat{V}_d(s)$. The theorem bounds the composite graph cost J. ∎

For reduced-graph certification, exact recursion caches (13) for every reachable tuple $(j, \mathbf{m}, \boldsymbol{\beta}, h)$ and enumerates every action in $\mathcal{U}_s(s)$. Exhaustive evaluation returns a minimum-cost task-completing sequence or $+\infty$ if none exists in the fixed sampled graph; a finite-horizon induction proof is included in the supplementary material. Exact optimality is established over the specified finite candidate graph.

With maximum branching factor $B_{\max}$ and at most H base-policy actions, one depth-d decision evaluation costs $O(HB_{\max}^{d+1})$ in the worst case; at most H replans give a direct whole-route bound of $O(H^2 B_{\max}^{d+1})$. Memoization reduces repeated evaluation when prefixes reach the same mask state. The Results section and Supplementary Table S2 quantify the cost–computation trade-off for $d = 1, \ldots, 6$.

V. Experimental Protocol

A. Fixed Geometry and Required-Target Profiles

The primary study uses the public ISS mesh distributed by the National Aeronautics and Space Administration (NASA) [34]. Its complete transformed geometry contains 247 525 triangles. 90 equal-represented-area samples define the original surface denominator; 41 are observable from at least one of the 24 retained candidate observations. The directed archive contains 24 initial-to-observation and 24×23 inter-observation records, of which 264 pass every enabled gate. Every method receives the same graph and scenario-specific available-node set.

Required-target profiles are selected deterministically before scenario evaluation by centroid-first, normalized farthest-point sampling within three equal-width mesh slabs. The primary nine-target profile assigns three samples to each slab; Supplementary Table S1 gives the complete 6-, 9-, and 12-target inventories. These synthetic profiles distribute prescribed samples across the observable station geometry.

The required set remains fixed when a scenario removes observation nodes. Losing every available view of a required item makes the task impossible in the library. Otherwise, a valid route or exhausted exact search establishes feasibility status; cases without either certificate remain unresolved.

B. Dynamics, Costs, and Observation Accounting

Each rest-to-rest SOOA lasts 90 s and uses fixed-step RK4 propagation with $\Delta t = 3\ s$. The HCW mean motion is 1.13137×10^{-3} rad/s, the acceleration limit is $u_{\max} = 0.060\ \mathrm{m/s^2}$, and the speed limit is $v_{\max} = 2.0\ m/s$. The finite-body radius is $r_{\mathrm{veh}} = 0.80\ m$, which conservatively bounds the chaser visual model, and the prescribed full-mesh clearance margin is $d_{\mathrm{safe}} = 2.0\ m$. Terminal position and velocity tolerances are 0.5 m and 0.05 m/s. For the reported simulations, we set $\lambda_v = 5$, $\lambda_e = 0.18$, $\lambda_s = 120$, and $\lambda_a = 0.05$, weighting maneuver increment, RMS distance to the destination, clearance violation, and per-edge action cost, respectively, in (8). The base-policy denominator floor is $\epsilon = 0.05$, and the action budget is 14. Required-only completion has no additional global area threshold.

Candidate visibility uses a rectangular $70^\circ \times 50^\circ$ FOV (horizontal $\times$ vertical), giving $\alpha_{\max}^{\mathrm{h}} = 35^\circ$ and $\alpha_{\max}^{\mathrm{v}} = 25^\circ$ in (3). The camera range is 2–54 m and the maximum incidence angle is 60°; full-mesh LOS is checked separately. Target-identity unions exclude duplicate credit. Reporting includes required-target completion $C^{\mathcal{K}}$, priority-weighted inspectable score C^{insp}, and unweighted fractions over the 41 observable and 90 original samples. Equal represented sample areas make the last fraction the sampled-area estimate; priority multipliers affect only the weighted score.

C. Study Chronology and Fresh Confirmation

An exploratory campaign evaluated three fixed profiles without feasibility rejection and informed the confirmation's comparator design. Its results are reported in Supplementary Table S3.

A fresh confirmation compares depth-three ADP with one-step ADP and local search initialized by a completed one-step ADP route. Methods, penalties, scenario identities, and implementation hashes were frozen before evaluating the new outcomes. The confirmation retains the nine prescribed targets and depth-three policy, with 12 validation, 50 test, and 30 shifted scenarios. Its seeds are

disjoint from all development runs, and every generated case is retained.

Each scenario samples a node-removal probability uniformly from $[0, 0.10]$, then removes candidates independently using that probability. It multiplies priorities for approximately 6%–14% of targets by a factor in $[2, 4]$. Shifted scenarios use removal probabilities in $[0.15, 0.25]$, priority fractions of 12%–20%, and factors in $[5, 8]$. Priorities affect background scores and the coverage-only ablation, but not the required-target goal or stage cost. All scenarios perturb one frozen graph, and distinct seeds may yield identical routes.

The post-confirmation depth diagnostic evaluated $d = 1, \ldots, 6$ on the 12 validation scenarios, distinct from the test and shifted sets. It assessed the previously fixed depth-three setting and supported its retention for the subsequent all-solvable benchmark. Its outcomes are presented first in Section A and Supplementary Table S2.

D. Matched Baselines and Statistical Reporting

The confirmation includes the task-aware greedy incumbent, one-step ADP, ADP-seeded local search, and depth-three ADP under identical tasks, dynamics, safety gates, and action budgets. One-step ADP uses (18) with $d = 1$. Local search applies three audited best-improvement reversal passes to its completed route and retains the feasible seed if improvement fails. This comparator improves an ADP-initialized ordering; it does not introduce new viewpoints. Depth-three ADP uses the original greedy base value without a one-step fallback.

A coverage-only ablation retains the earlier 80% priority-weighted stopping rule and is evaluated against the same required-target identities. It tests the mission objective separately from the matched complete-task cost comparison.

Success and missing-target counts use every scenario; graph cost and physical Δv use jointly completed ADP/seeded-local pairs. For a route with final covered-target mask $\mathbf{m}$, the all-scenario score is $J_{\mathrm{pen}} = J + 500(1 - C^{\mathcal{K}}(\mathbf{m})) + 500(1 - \Gamma_{\mathcal{K}}(\mathbf{m}))$, with fixed evaluation penalties distinct from physical maneuver cost. Paired-mean percentile intervals use 10 000 resamples, and exact two-sided sign tests exclude numerical ties. Timing includes initialization, including the one-step seed for local search; safety-screen counts cover the selected planner call.

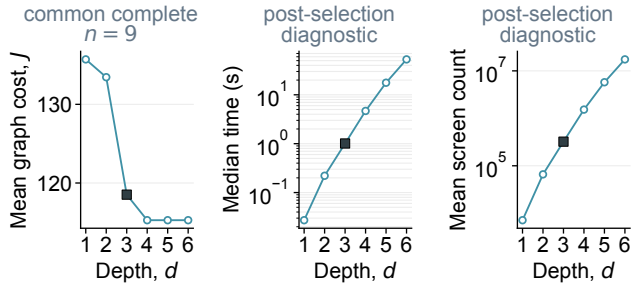
E. All-Solvable Redundant-Viewpoint Benchmark

After the depth diagnostic, a redundant-viewpoint pilot tested whether additional viewpoints reduce the visibility bottleneck while permitting independent route certification. Three frozen seeds generated 33-view graphs on the same mesh, with the same nine required targets, initial state, camera model, dynamics, costs, and safety limits. Each graph contained three separated views per required target and six connecting views anchored to other observable samples. Candidate positions were sampled 26–42 m from their source target within a 35° cone around its surface normal, with the camera aimed at that target. From 64 attempts per source, deterministic first-valid and farthest-point selection retained geometry-valid views separated by at least 4 m. All rejected attempts were archived. Each graph's $33 + 33 \times 32 = 1\,089$ directed transfers were regenerated and audited against the full mesh.

Each graph supplied one reference case, ten nominal cases and ten shifted cases, using the confirmation's node-removal probability ranges. All 63 cases were retained. The 59 cases with an independent feasible certificate define the primary all-solvable benchmark (three reference, 30 nominal and 26 shifted cases). The four remaining shifted cases form a separate visibility-loss stress test. Background weights were fixed because they do not affect required-only completion or graph cost. The generation rules, seeds and source hashes were frozen before graph construction. Developed after analysis of the confirmation failures, this benchmark provides development evidence on viewpoint redundancy.

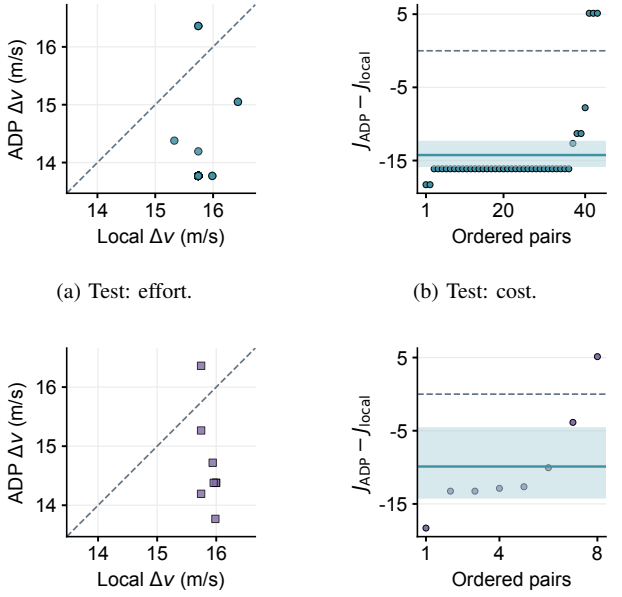
An independent mixed-integer linear program (MILP) served as an offline finite-graph reference and feasibility certificate. Binary arc and node variables enforced flow, required-target coverage, no revisits and the 14-observation budget; increasing node ranks excluded disconnected cycles. Its objective was the same additive graph cost, including the 0.05 action charge. A reconstructed valid route certified feasibility. Missing required visibility or a solver infeasibility result certified impossibility in the available graph; a time limit without a valid route remained unresolved. We labelled a case proven optimal only when HiGHS supplied an optimality proof and the reconstructed route cost matched its branch-and-bound lower bound within numerical tolerance. A time-limited feasible route was retained as a checked incumbent, not an optimum. Certification preceded ADP evaluation and supplied no route to either ADP depth.

Greedy construction, one-step ADP, depth-three ADP and the independent MILP each received a nominal 15 s budget, including initialization or model construction and excluding shared graph generation. The pilot used Python 3.12.3 and SciPy 1.11.4/HiGHS on Ubuntu. Each method–case pair was timed once; greedy/ADP order was shuffled, while MILP timing came from the preceding certification phase. The MILP timing quantifies the computational cost of the finite-graph reference search. Native solver termination could slightly exceed its requested limit. Planner success is evaluated on the 59 independently certified solvable cases; the four visibility-infeasible cases are reported separately. Cost gaps used only cases with proven optima.



(a) Mean validation cost. (b) Median planning time. (c) Safety-screen evaluations.

Fig. 3. Post-confirmation depth diagnostic on 12 validation cases. The square marks the retained depth-three setting; cost uses the nine cases completed at every depth, and panels (b), (c) use logarithmic scales.



(a) Test: effort. (b) Test: cost.

(c) Shifted: effort. (d) Shifted: cost.

Fig. 4. ADP/seeded-local comparison on 43 jointly completed test pairs (a), (b) and 8 shifted pairs (c), (d). Panels (a), (c) show physical Δv; panels (b), (d) show ordered graph-cost differences. Dashed lines denote parity; solid lines and bands give means and 95% paired-bootstrap intervals.

VI. Results

A. Rollout Depth and Computational Cost

The separate validation depth study evaluated $d = 1, \ldots, 6$ on 12 scenarios, distinct from the fixed-library test and shifted sets. All depths completed the same nine tasks; the other three cases lacked required visibility. Depth three reduced mean graph cost by 11.19% relative to depth two. Depth four reduced it by a further 2.74% but required 4.63 times the median planning time, whereas depths five and six produced no further cost reduction. These results support depth three as a practical balance between graph cost and computation; this setting was retained for the subsequent all-solvable benchmark. Fig. 3 summarizes this trade-off; Supplementary Table S2 gives the complete values and timing protocol.

B. Complete Required-Target Inspection

Depth-three ADP completed all nine targets in 43/50 test scenarios and 9/30 shifted scenarios. Seeded local search completed 43/50 and 8/30, respectively. Table II summarizes these confirmation results. Every completed mission achieved $C^{\mathcal{K}} = 100\%$, with optional background coverage reported separately.

Every confirmation case not completed by depth-three ADP had lost all available views of at least one required target. Depth-three ADP therefore completed all 43 test and all 9 shifted cases that retained required visibility; each returned route supplied a feasibility witness. Five required targets had only one candidate view, making the fixed library sensitive to node removal.

C. ADP Policy Improvement at Matched Completion

The predeclared ADP-minus-seeded-local penalized endpoint had mean differences of -12.256 on the 50 test scenarios and -31.999 on the 30 shifted scenarios. Their 95% intervals were $[-14.155, -10.212]$ and $[-91.494, -1.241]$, respectively. Physical-resource comparisons below use the jointly completed missions.

Fig. 4 compares the 43 jointly completed test pairs and 8 shifted pairs. Mean ADP-minus-local graph-cost differences were -14.251 and -9.890, respectively. Table III reports the paired uncertainty and win/tie/loss counts.

On the 43 completed test pairs, mean Δv decreased from 15.777 m/s to 14.033 m/s, an 11.05% reduction. The paired reduction was 1.744 m/s, with a 95% interval of 1.525–1.923 m/s; ADP won 40 pairs and lost three. The eight shifted pairs had a mean reduction of 1.207 m/s. Its bootstrap interval excluded zero, while the exact sign test gave $p = 0.0703$.

The confirmation quantifies improvements over one-step ADP and ADP-seeded local search. Finite-graph unit tests exercise the base-cost bound, connecting actions, and exact recursion.

D. Objective Ablation and Safety

To test whether aggregate coverage can substitute for prescribed-target completion, we reran the same scenarios with the earlier coverage-only stopping rule, $C^{\text{insp}} \geq 0.80$, while retaining the same nine required targets for evaluation. The coverage-only policy completed all nine required targets in only 1/50 test and 0/30 shifted scenarios. By contrast, 41 of 43 completed depth-three ADP test missions finished below the 80% priority-weighted coverage threshold while accepting all nine prescribed items; mean background coverage over those 43 successes was 72.32%.

TABLE II

Required-target results. Resource means use successful routes of each method; the final column includes every scenario with the fixed failure penalty

Split	Method	Complete	Mean SOOAs	Mean Δv (m/s)	Mean J	Penalized J
Test	Task-aware incumbent	0/50	–	–	–	911.242
Test	One-step ADP	43/50	8.21	16.512	135.945	256.913
Test	ADP-seeded local search	43/50	8.21	15.777	129.892	251.707
Test	Depth-three ADP	43/50	9.00	14.033	115.641	239.451
Shifted	Task-aware incumbent	0/30	–	–	–	851.059
Shifted	One-step ADP	8/30	9.25	17.012	140.102	770.694
Shifted	ADP-seeded local search	8/30	8.88	15.888	130.841	768.224
Shifted	Depth-three ADP	9/30	9.11	14.656	120.751	736.225

TABLE III

ADP-minus-seeded-local effects. The predeclared penalized endpoint includes all scenarios; physical resources and graph cost use jointly complete pairs. Intervals are percentile 95% paired-bootstrap intervals; sign tests exclude ties

Split	Endpoint	Pairs	Mean difference	95% interval	Win/tie/loss	Sign-test p
Test	J_{pen} (all)	50	-12.256	$[-14.155, -10.212]$	40/7/3	3.02×10^{-9}
Test	J	43	-14.251	$[-15.715, -12.461]$	40/0/3	3.02×10^{-9}
Test	Δv (m/s)	43	-1.744	$[-1.923, -1.525]$	40/0/3	3.02×10^{-9}
Shifted	J_{pen} (all)	30	-31.999	$[-91.494, -1.241]$	8/21/1	0.0391
Shifted	J	8	-9.890	$[-14.111, -4.688]$	7/0/1	0.0703
Shifted	Δv (m/s)	8	-1.207	$[-1.713, -0.575]$	7/0/1	0.0703

Full-mesh occlusion and collision checks include structure outside the required target set; accepted edges satisfy the enabled clearance and input audits.

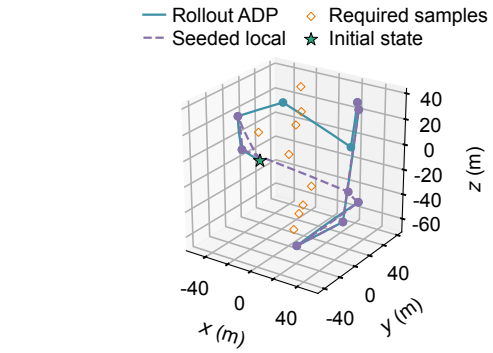
E. Representative Required-Target Route

Fig. 5 shows the selected routes, target completion, and accumulated Δv for the representative test pair.

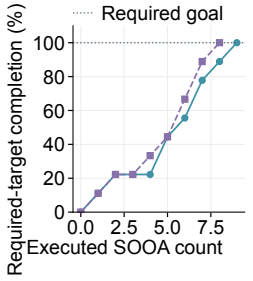
The representative test case is the jointly completed scenario nearest the median ADP-minus-local graph-cost difference. Ties are resolved deterministically using the scenario identifier. Both routes complete all 9 prescribed targets. Rollout ADP uses 9 SOOAs, $J = 113.472$, and $\Delta v = 13.769$ m/s; local search uses 8 SOOAs, $J = 129.605$, and $\Delta v = 15.744$ m/s. The respective background inspectable scores are 77.22% and 64.97%. Repropagation reproduces both archived costs and velocity increments to numerical tolerance.

F. All-Solvable Benchmark Results

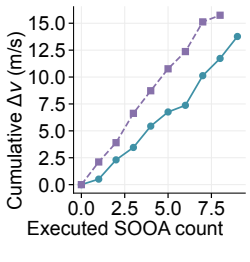
The three new graphs supplied at least three valid views of every required target, with 413, 439 and 386 admissible directed edges. Graph construction took 28.7–30.2 s per graph. Independent certification found all three reference cases, all 30 nominal cases and 26 of 30 shifted cases feasible. The other four shifted cases lacked any available view of a required target; none remained unresolved. We therefore use the 59 certified-solvable cases as the primary benchmark for planner success rates and retain the four cases in a separate visibility-loss stress test.



(a) Generated LVLH routes.



(b) Required-target completion.



(c) Cumulative physical Δv.

Fig. 5. Representative required-target test pair selected by the median-effect rule. Both methods complete the same required set. Markers identify terminal observations and prescribed samples; the star denotes the common initial state.

Table IV reports completion on the certified-solvable cases. Depth-three ADP completed all 59, one-step ADP missed two shifted cases, and greedy construction completed 33. Both ADP depths stayed within the cost of

TABLE IV

Primary all-solvable development benchmark. Each denominator is independently certified feasible; four additional shifted cases failing the visibility certificate are retained as a separate stress test. The independent MILP row supplies the feasibility certificate, while its proven-optimal subset provides the finite-graph cost reference.

Method	Reference (3)	Nominal (30)	Shifted (26)	All-solvable (59)
Task-aware incumbent	1/3	13/30	19/26	33/59
One-step ADP	3/3	30/30	24/26	57/59
Depth-three ADP	3/3	30/30	26/26	59/59
Independent MILP	3/3	30/30	26/26	59/59

every completing greedy route, giving 66 method–case checks.

The MILP proved minimum graph costs for nine nominal and 18 shifted cases, providing an exact reference on those finite graphs. Depth-three ADP had a mean relative gap of 0.96% across these 27 cases, while matching the MILP completion rate on all 59 independently feasible cases. The other 32 feasible MILP runs reached the time limit with checked routes but unproven optimality, so those routes were not treated as optima. Median nominal/shifted times were 0.001/0.001 s for greedy construction, 0.046/0.037 s for one-step ADP, 4.173/2.085 s for depth-three ADP, and 15.007/5.373 s for the MILP. Thus, the proposed ADP traded a small gap to the proven finite-graph optima for shorter median computation, while reducing mean cost relative to one-step ADP by 5.493 units across 30 nominal pairs and 7.012 units across 24 shifted pairs.

Post-run reconstruction verified all 252 method–case records and 3 267 numeric edge records. The nominal splits contain seven, nine, and ten distinct availability masks for the three graphs, respectively.

G. Required-Target ROS Execution

ROS-native HCW dynamics and linear-quadratic regulator control track a fixed ADP reference without online replanning. An observation is accepted only when position error is at most 0.5 m and speed is at most 0.05 m/s. The twelve-view hybrid execution also requires 95% weighted background coverage. It uses 20-Hz control and state publication, 0.05 s integration, a $0.060\ \mathrm{m/s^2}$ acceleration limit, a 1.50 m/s speed limit, and the 2 m full-mesh safety margin.

The normal-speed ROS 2 Jazzy run accepted all twelve observations at 1800.035 s, covering 9/9 required targets and 95.86% of weighted background samples. Maximum terminal position error and speed were 0.050 m and 0.0038 m/s. Executed Δv was 20.477 m/s, versus 18.027 m/s for the transfer-and-hold reference. The full-mesh audit found no surface crossing; minimum swept-body clearance was 3.380 m above the 2 m margin, and peak acceleration was $0.0600\ \mathrm{m/s^2}$. Supplementary Section S5 reports viewpoint refinement, message counts, synchronization, and timing diagnostics.

Fig. 6 illustrates reference tracking and accepted observations using synchronized camera frames.

Fig. 7 shows a separate repeat execution of the same reference (Supplementary Video 1).

VII. Discussion

The results support model-based rollout ADP for prescribed-target inspection: a base-policy completion supplies a checkable continuation, and Bellman lookahead improves the next action. Required targets determine completion, while an optional background threshold can be added through the same terminal predicate. This sequencing layer complements learning-based proximity control [24], [25], [28]. An infinite rollout value indicates that the evaluated prefixes and base-policy continuations yielded no completion certificate. Graph infeasibility requires an independent certificate.

The fixed-library comparison measures maneuver savings against one-step ADP and seeded reversal search. The independent MILP supplies feasibility certificates and exact cost references for its proven-optimal subset. Under its 15-s budget, 32 feasible cases remained without optimality proofs, and its median runtimes exceeded those of depth-three ADP. The proposed method therefore targets the speed-quality trade-off: it matched MILP completion on the 59 certified-solvable cases. Its mean relative graph-cost gap was 0.96% across the 27 cases with proven optima.

The development benchmark uses three graphs on one station geometry, distinct from the confirmation's fixed viewpoint library. Its availability scenarios are dependent, and each method–case pair was timed once. Broader validation should examine additional geometries, independently generated missions, and repeated runtime measurements.

VIII. Conclusion

This paper developed viability-preserving rollout ADP for prescribed-target inspection over audited orbital arcs. It completed every feasible fixed-library confirmation case and reduced mean Δv by 11.05% relative to seeded local search on 43 jointly completed test missions. In the redundant-viewpoint development benchmark, depth-three ADP completed all 59 cases independently certified as solvable by the MILP, with shorter median runtimes than the MILP. Its mean relative graph-cost gap was 0.96% across the 27 cases with proven optima. These results support completion-certified sequencing as a computationally lighter alternative to exact finite-graph search, within the stated graph and audit assumptions, and motivate a larger frozen evaluation of viewpoint redundancy.

REFERENCES

[1] A. Flores-Abad, O. Ma, K. Pham, and S. Ulrich, "A review of space robotics technologies for on-orbit servicing," *Progress in*

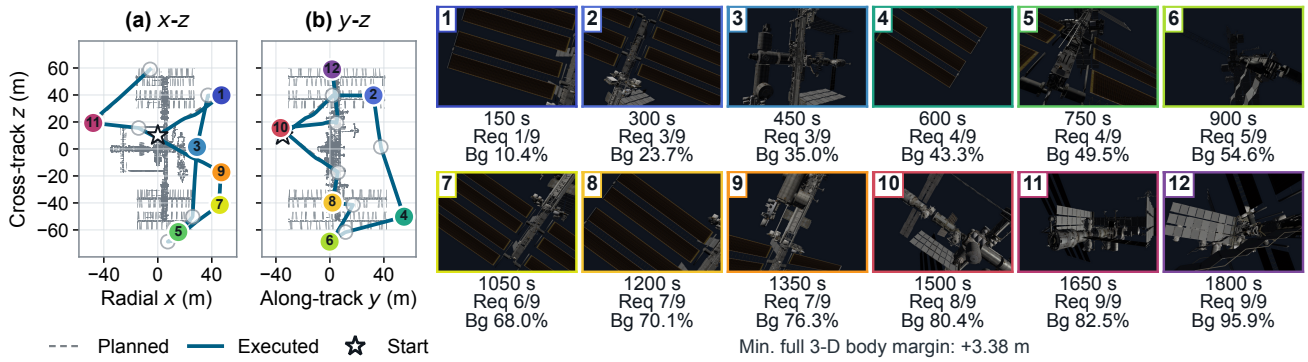
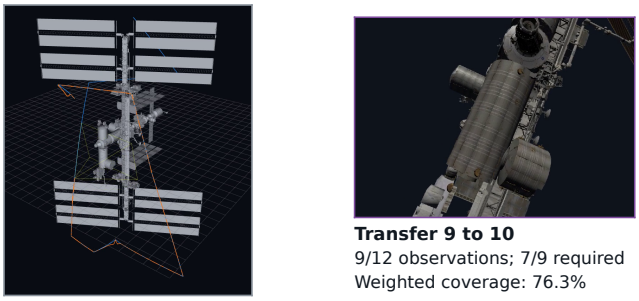


Fig. 6. Twelve-view ROS 2 execution with 90 s transfers and 60 s settling intervals. Panels (a), (b) highlight odd and even observations at equal scale; the camera rows show all accepted views with mission time, required targets (Req), and weighted coverage (Bg). The run accepts 9/9 required targets and 95.86% background coverage while maintaining the full-mesh safety margin.



(a) Global trajectory. (b) Onboard camera.

Fig. 7. RViz display: (a) planned (blue) and executed (orange) trajectories with camera FOV (yellow); (b) onboard Gazebo camera. Crops from one frame during transfer 9–10.

Aerospace Sciences, vol. 68, pp. 1–26, 2014.

[2] S. Kimura, T. Okuyama, N. Yoshioka, and Y. Wakabayashi, "Robot-aided remote inspection experiment on STS-85," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 36, no. 4, pp. 1290–1297, 2000.

[3] S. Faghihi, S. Tavana, and A. H. J. de Ruiter, "Kinodynamic on-orbit inspection path planning for full-coverage inspection in close proximity of space structures," *Acta Astronautica*, vol. 198, pp. 354–365, 2022.

[4] S. Faghihi, S. Tavana, and A. H. J. de Ruiter, "Multiple spacecraft coordination and motion planning for full-coverage inspection of large complex space structures," *Acta Astronautica*, vol. 202, pp. 119–129, 2023.

[5] B. Apodaca, T. Helgeson, E. Atkins, and L. Stirling, "In-orbit space structure inspection trajectory generation," *IEEE Robotics and Automation Letters*, vol. 10, no. 9, pp. 8810–8817, 2025.

[6] W. R. Scott, G. Roth, and J.-F. Rivest, "View planning for automated three-dimensional object reconstruction and inspection," *ACM Computing Surveys*, vol. 35, no. 1, pp. 64–96, 2003.

[7] Z. Hou, B. Jiao, and Z. Dang, "Relative orbit design of CubeSats for on-orbit visual inspection of China Space Station," *Advances in Space Research*, vol. 73, no. 1, pp. 170–186, 2024.

[8] T. Deka and J. W. McMahon, "Astrodynamics-informed kinodynamic sampling-based motion planning for relative spacecraft motion," *Journal of Guidance, Control, and Dynamics*, vol. 46, no. 12, pp. 2330–2345, 2023.

[9] D. P. Bertsekas, J. N. Tsitsiklis, and C. Wu, "Rollout algorithms for combinatorial optimization," *Journal of Heuristics*, vol. 3, no. 3, pp. 245–262, 1997.

[10] D. P. Bertsekas, *Rollout, Policy Iteration, and Distributed Reinforcement Learning*. Athena Scientific, 2020.

[11] B. E. Tweddle, T. P. Setterfield, A. Saenz-Otero, and D. W. Miller, "An open research facility for vision-based navigation onboard the International Space Station," *Journal of Field Robotics*, vol. 33, no. 2, pp. 157–186, 2016.

[12] K. Wang, T. Meng, W. Wang, and J. Lei, "Control barrier function based trajectory generation and tracking control for spacecraft inspection mission under multiple safety constraints," *Advances in Space Research*, vol. 73, no. 3, pp. 2080–2097, 2024.

[13] W. H. Clohessy and R. S. Wiltshire, "Terminal guidance system for satellite rendezvous," *Journal of the Aerospace Sciences*, vol. 27, no. 9, pp. 653–658, 1960.

[14] J. R. Wertz, D. F. Everett, and J. J. Puschell, Eds., *Space Mission Engineering: The New SMAD*. Microcosm Press, 2011.

[15] T. Williams, "Orbital inspection vehicle trajectories based on line-of-sight maneuvers," *Acta Astronautica*, vol. 50, no. 1, pp. 49–53, 2002.

[16] H. Zhou, B. Jiao, Z. Hou, and Z. Dang, "Satellite formation control for cooperative observation around the rotating China Space Station," *Space: Science & Technology*, vol. 6, p. 0348, 2026.

[17] S. M. LaValle, "Rapidly-exploring random trees: A new tool for path planning," Department of Computer Science, Iowa State University, Tech. Rep. TR 98-11, October 1998.

[18] S. Karaman and E. Frazzoli, "Sampling-based algorithms for optimal motion planning," *The International Journal of Robotics Research*, vol. 30, no. 7, pp. 846–894, 2011.

[19] M. Maestrini and P. Di Lizia, "Guidance strategy for autonomous inspection of unknown non-cooperative resident space objects," *Journal of Guidance, Control, and Dynamics*, vol. 45, no. 6, pp. 1126–1136, 2022.

[20] J. Zhou, G. Kang, J. Wu, X. Tao, C. Xu, Y. Qiu, and J. Wu, "Tomographic inspection path planning for spacecraft with complex structures," *Acta Astronautica*, vol. 239, pp. 162–183, 2026.

[21] D. P. Bertsekas and J. N. Tsitsiklis, *Neuro-Dynamic Programming*. Athena Scientific, 1996.

[22] D. P. Bertsekas, "Dynamic programming and suboptimal control: A survey from ADP to MPC," *European Journal of Control*, vol. 11, no. 4–5, pp. 310–334, 2005.

[23] D. P. Bertsekas, "Value and policy iterations in optimal control and adaptive dynamic programming," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 28, no. 3, pp. 500–509, 2017.

[24] Q. Hu, H. Yang, H. Dong, and X. Zhao, "Learning-based 6-DOF control for autonomous proximity operations under motion constraints," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 57, no. 6, pp. 4097–4109, 2021.

[25] Y. Tian, Y. Shi, X. Shao, Q. Hu, H. Yang, and Z. H. Zhu, "Spacecraft proximity operations under motion and input constraints: A learning-based robust optimal control approach," *IEEE*

Transactions on Aerospace and Electronic Systems, vol. 60, no. 6, pp. 7838–7852, 2024.

[26] C. Luo, X. Ning, and R. Tang, "Online optimal attitude stabilization via reinforcement learning for rigid spacecraft with dynamic uncertainty," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 4, pp. 10 471–10 482, 2025.

[27] R. Tang, C. Luo, T. Wang, X. Ning, and C.-Y. Wen, "Reach-avoid differential graphical games for single evader and multiple pursuers with nonlinear dynamics," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 24 545–24 558, 2025.

[28] D. van Wijk, K. Dunlap, M. Majji, and K. Hobbs, "Safe spacecraft inspection via deep reinforcement learning and discrete control barrier functions," *Journal of Aerospace Information Systems*, vol. 21, no. 12, pp. 996–1013, 2024.

[29] J. Achiam, D. Held, A. Tamar, and P. Abbeel, "Constrained policy optimization," in *Proceedings of the 34th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 70, 2017, pp. 22–31.

[30] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu, "Safe reinforcement learning via shielding," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018, pp. 2669–2678.

[31] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs for safety critical systems," *IEEE Transactions on Automatic Control*, vol. 62, no. 8, pp. 3861–3876, 2017.

[32] K. P. Wabersich and M. N. Zeilinger, "Predictive control barrier functions: Enhanced safety mechanisms for learning-based control," *IEEE Transactions on Automatic Control*, vol. 68, no. 5, pp. 2638–2651, 2023.

[33] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher, "An analysis of approximations for maximizing submodular set functions—I," *Mathematical Programming*, vol. 14, no. 1, pp. 265–294, 1978.

[34] National Aeronautics and Space Administration, "International Space Station glTF Model," https://science.nasa.gov/resource/international-space-station-3d-model/, 2019, updated Nov. 2, 2024; accessed Aug. 10, 2026.